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Meta Dynamic Pricing: Transfer Learning Across Experiments

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Abstract. We study the problem of learning shared structure *across* a sequence of dynamic pricing experiments for related products. We consider a practical formulation in which the unknown demand parameters for each product come from an unknown distribution (prior) that is shared across products. We then propose a meta dynamic pricing algorithm that learns this prior online while solving a sequence of Thompson sampling pricing experiments (each with horizon T) for N different products. Our algorithm addresses two challenges: (i) balancing the need to learn the prior (*meta-exploration*) with the need to leverage the estimated prior to achieve good performance (*meta-exploitation*) and (ii) accounting for uncertainty in the estimated prior by appropriately "widening" the estimated prior as a function of its estimation error. We introduce a novel prior alignment technique to analyze the regret of Thompson sampling with a misspecified prior, which may be of independent interest. Unlike prior-independent approaches, our algorithm's meta regret grows sublinearly in N, demonstrating that the price of an unknown prior in Thompson sampling can be negligible in experiment-rich environments (large N). Numerical experiments on synthetic and real auto loan data demonstrate that our algorithm significantly speeds up learning compared with prior-independent algorithms.

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1. Introduction

Experimentation is popular on online platforms to optimize a wide variety of elements, such as search engine design, homepage promotions, and product pricing. This has led firms to perform an increasing number of experiments, and several platforms have emerged to provide the infrastructure for these firms to perform experiments at scale (see, e.g., Optimizely 2019). State-of-the-art techniques in these settings employ bandit algorithms (e.g., Thompson sampling), which seek to adaptively learn treatment effects while optimizing performance *within* each experiment (Thompson 1933, Scott 2015). However, the large number of related experiments begs the question: can we transfer knowledge *across* experiments?

We study this question for Thompson sampling algorithms in dynamic pricing applications that involve a large number of related products. Dynamic pricing algorithms enable retailers to optimize profits by sequentially experimenting with product prices and learning the resulting customer demand (Kleinberg and Leighton 2003, Besbes and Zeevi 2009). Such algorithms are shown to be especially useful for products that exhibit relatively short life cycles (Ferreira et al. 2015), stringent inventory constraints (Xu et al. 2019), strong competitive effects (Fisher et al. 2021), or the ability to offer personalized coupons/pricing (Ban and Keskin 2021, Zhang et al. 2017). In all these cases, the demand of a product is estimated as a function of the product's price (chosen by the decision maker) and a combination of exogenous features as well as product- and customer-specific features. Through carefully chosen price experimentation, the decision maker can learn the price-dependent demand function for a given product and choose an optimal price to maximize profits (Cohen et al. 2020, Qiang and Bayati 2016, Javanmard and Nazerzadeh 2019). Dynamic pricing algorithms based on Thompson sampling are shown to be particularly successful in striking the right balance between exploring (learning the demand) and exploiting (offering the estimated optimal price) and are widely considered to be the state of the art (Thompson 1933, Agrawal and Goyal 2013, Russo and Van Roy 2014, Ferreira et al. 2018).

The decision maker typically runs a separate pricing experiment (i.e., dynamic pricing algorithm) for each product (or for a set of simultaneously offered products). However, this approach can waste valuable samples rediscovering information that could have been learned from previously offered related products. For example, students may be more price-sensitive than general customers; as a result, many firms, such as restaurants, retailers, and movie theaters, offer student discounts. This implies that the coefficient of student-specific price elasticity in the demand function is positive for many products (although the specific value of the coefficient likely varies across products). Similarly, winter clothing may have higher demand in the fall and lower demand at the end of winter. This implies that the demand functions of winter clothing may have similar coefficients for the features indicating time of year. In general, there may even be complex correlations between coefficients of the demand functions of products that are shared. For example, the price-elasticities of products are often negatively correlated with their demands; that is, customers are willing to pay higher prices when the demand for a product is high. When offering *multiple* products simultaneously, one must additionally learn cross-product price elasticities in the demand function (to model substitution effects), which may also exhibit patterns that can be learned from substitution patterns of related products in historical data. For example, substitution effects may be stronger between more similar products or among more price-sensitive customers, such as students.

Thus, one may expect that the demand functions for related products may share some (a priori unknown) common structure, which can be learned across products. Note that the demand functions are unlikely to be exactly the same, so a decision maker would still need to conduct separate pricing experiments for each product. However, accounting for shared structure during these experiments may significantly speed up learning per product (or per set of products if offering multiple products simultaneously), thereby improving profits.

In this paper, we propose an approach to learn a shared structure across pricing experiments. We begin by noting that the key (and only) design decision in Thompson sampling methods is the Bayesian prior over the unknown parameters. This prior captures a shared structure of the kind we describe—for example, the mean of the prior on the student-specific price-elasticity coefficient may be positive with a small standard deviation. It is well known that choosing a good (bad) prior significantly improves (hurts) the empirical performance of the algorithm (Chapelle and Li 2011, Honda and Takemura 2014, Liu and Li 2015, Russo et al. 2018). However, the prior is typically unknown in practice, particularly when the decision maker faces a cold start. Although the decision maker can use a *prior-independent* algorithm (Agrawal and Goyal 2013), such an approach achieves poor empirical performance because of over-exploration; we demonstrate a substantial gap between the prior-independent and -dependent approaches in our experiments on synthetic and real data. In particular, knowledge of the correct prior enables Thompson sampling to appropriately balance exploration and exploitation (Russo and Van Roy 2014). Thus, the decision maker needs to learn the true prior (i.e., shared structure) across products to achieve good performance. We propose a meta dynamic pricing algorithm that efficiently achieves this goal.

We first formulate the problem of learning the true prior online while solving a sequence of pricing experiments for different products. Our meta dynamic pricing algorithm requires two key ingredients. First, for each product, we must balance the need to learn about the prior ("meta-exploration") with the need to leverage the prior to achieve strong performance for the current product ("meta-exploitation"). In other words, our algorithm balances an additional exploration–exploitation trade-off across price experiments. Second, a key technical challenge is that finite-sample estimation errors of the prior may significantly impact the performance of Thompson sampling for any given product. In particular, vanilla Thompson sampling may fail to converge with an incorrect prior; as a result, directly using the estimated prior across products can result in poor performance. To this end, we introduce a novel "prior alignment" technique to analyze the regret of Thompson sampling with a misspecified prior, which may be of independent interest.

Using our alignment technique, we show surprisingly that *despite* prior misspecification, greedy updating of the prior is sufficient to learn effectively across pricing experiments when the prior covariance is known. However, when the prior has an unknown covariance matrix, it is beneficial to widen the estimated prior covariance by a term that is a function of the prior's estimated finite-sample error. Thus, we use a more conservative approach (a wide prior) for earlier products when the prior is uncertain; over time, we gain a better estimate of the prior and can leverage this knowledge for better empirical performance. Our algorithm provides an exact prior correction path over time to achieve strong performance guarantees across all pricing problems. We prove that, when using our algorithm, the price of an unknown prior for Thompson sampling is negligible in experiment-rich environments (i.e., as the number of products grows large).

1.1. Related Literature

Experimentation is widely used to optimize decisions in a data-driven manner. This has led to a rich literature on bandits and A/B testing (Lai and Robbins 1985, Auer 2002, Dani et al. 2008, Rusmevichientong and Tsitsiklis 2010, Besbes et al. 2014, Johari et al. 2015, Bhat et al. 2019). This literature primarily proposes learning

algorithms for a single experiment, and our focus is on meta-learning across experiments. Meta-learning can take the form of constructing an empirical Bayesian prior (Raina et al. 2006, Anderer et al. 2019), data pooling (Gupta and Kallus 2020), or leveraging low-dimensional structure between problems (Bastani 2021). We take an empirical Bayesian approach to sequential decision making. Although there has been some prior work on meta-learning in bandits (Hartland et al. 2006, Maes et al. 2012, Wang et al. 2018, Sharaf and Daumé 2019) and more generally in reinforcement learning (Finn et al. 2017, 2018; Yoon et al. 2018), these papers only provide heuristics for learning exploration strategies given a fixed set of past problem instances. They do not prove any theoretical guarantees on the performance or regret of the meta-learning algorithm. To the best of our knowledge, our paper is the first to propose a meta-learning algorithm in a bandit setting with provable regret guarantees.

We study the specific case of dynamic pricing, which aims to learn an unknown demand curve in order to optimize profits. We focus on dynamic pricing because meta-learning is particularly important in this application; for example, online retailers, such as Rue La La, may run numerous pricing experiments for related fashion products. We believe that a similar approach could be applied to multiarmed or contextual bandit problems in order to inform the prior for Thompson sampling across a sequence of related bandit problems.

Dynamic pricing has been found to be especially useful in settings with short life cycles or limited inventory (e.g., fast fashion or concert tickets; see Ferreira et al. 2015, Xu et al. 2019), among online retailers that constantly monitor competitor prices and adjust their own prices in response (Fisher et al. 2017), or when prices can be personalized based on customer-specific price elasticities (e.g., through personalized coupons; see Zhang et al. 2017). Several papers have designed near-optimal dynamic pricing algorithms for pricing a product by balancing the resulting exploration–exploitation trade-off (Kleinberg and Leighton 2003, Araman and Caldentey 2009, Besbes and Zeevi 2009, Farias and Van Roy 2010, Broder and Rusmevichientong 2012, Harrison et al. 2012, den Boer and Zwart 2013, Keskin and Zeevi 2014). Recently, this literature has shifted focus to pricing policies that dynamically optimize the offered price with respect to exogenous features (Cohen et al. 2020, Qiang and Bayati 2016, Javanmard and Nazerzadeh 2019) and customer-specific price elasticity (Ban and Keskin 2021, Elmachtoub et al. 2021). We adopt the linear demand model proposed by Ban and Keskin (2021), which allows for feature-dependent heterogeneous price elasticities.

When sellers offer multiple products simultaneously, one may wish to perform price experiments *jointly* on a set of products to capture substitution effects or overlapping inventory constraints (Agrawal and Devanur 2014, Keskin and Zeevi 2014, Ferreira et al. 2018). However, in these papers, price experimentation is still performed independently on the current set of products, and any learned parameter knowledge is not shared across future sets of products to inform future demand learning. In contrast, we propose a meta dynamic pricing algorithm that learns the distribution of unknown parameters of the demand function across products. Although we focus largely on the single-product setting for ease of exposition, we show how our algorithm and theoretical results carry over straightforwardly for multiproduct settings with substitution effects; in fact, transfer learning from historical data may be even more valuable in these settings because the number of parameters (e.g., cross-product elasticities) to learn is much larger.

Our learning strategy is based on Thompson sampling, which is widely considered to be the state of the art for balancing the exploration–exploitation trade-off (Thompson 1933). Several papers study the sensitivity of Thompson sampling to prior misspecification. For example, Honda and Takemura (2014) show that Thompson sampling still achieves the optimal theoretical guarantee with an incorrect but uninformative prior but can fail to converge if the prior is not sufficiently conservative. Liu and Li (2015) provide further support for this finding by showing that the performance of Thompson sampling for any given problem instance depends on the probability mass (under the provided prior) placed on the underlying parameter; thus, one may expect that Thompson sampling with a more conservative prior (i.e., one that places nontrivial probability mass on a wider range of parameters) is more likely to converge when the true prior is unknown. It is worth noting that Agrawal and Goyal (2013) and Bubeck and Liu (2013) propose a prior-independent form of Thompson sampling, which is guaranteed to converge to the optimal policy even when the prior is unknown by conservatively increasing the variance of the posterior over time. However, the use of a more conservative prior creates a significant cost in empirical performance (Chapelle and Li 2011). For instance, Bastani et al. (2021) empirically find through simulations that the conservative prior-independent Thompson sampling is significantly outperformed by vanilla Thompson sampling *even* when the prior is misspecified.[1] We empirically find, through experiments on synthetic and real data sets, that learning and leveraging the prior can yield much better performance compared with a prior-independent approach. As such, the choice of prior remains an important design choice in the implementation of Thompson sampling (Russo et al. 2018). We propose a meta-learning algorithm that learns the prior across pricing experiments on related products to attain better performance. We also

empirically demonstrate that a naive approach of greedily using the updated prior performs poorly when the prior covariance is unknown because it may cause Thompson sampling to fail to converge to the optimal policy for some products. Instead, our algorithm gracefully tunes the width of the estimated prior as a function of the uncertainty in the estimate over time.

1.2. Main Contributions

We highlight our main contributions:

1. Model: We formulate our problem as a sequence of N different dynamic pricing problems, each with horizon T. Importantly, the unknown parameters of the demand function for each product are drawn independent and identically distributed (i.i.d.) from a shared (unknown) multivariate Gaussian prior.

2. Algorithm: We propose two meta-learning pricing policies, Meta-DP and Meta-DP++. The former learns only the mean of the prior, and the latter learns both the mean and the covariance of the prior across products. Both algorithms balance the need to learn the prior (meta-exploration) with the need to leverage the current estimate of the prior to achieve good performance (meta-exploitation). Meta-DP++ additionally accounts for uncertainty in the estimated prior by conservatively widening the prior as a function of its estimation error.

3. Theory: Unlike standard approaches, our algorithm can leverage shared structure across products to achieve regret that scales sublinearly in the number of products N. We prove upper bounds $\tilde{O}(d^2\sqrt{NT} + d^3\sqrt{T})$ and $\tilde{O}(\min\{d^2NT^{\frac{1}{2}}, d^4N^{\frac{1}{2}}T^{\frac{3}{2}}\}) = \tilde{O}(d^3(NT)^{\frac{2}{6}})$ on the meta regret of Meta-DP and Meta-DP++, respectively. In both cases, our meta-learning approach matches the performance of prior-independent algorithms for small N and outperforms them in experiment-rich experiments (i.e., when $N = \tilde{\Omega}(d)$ and $N = \tilde{\Omega}(d^4T^2)$, respectively). A key ingredient of our analysis is a prior alignment proof technique that may be of general interest for analyzing the regret of misspecified Thompson sampling instances.

4. Numerical experiments: We demonstrate on both synthetic and real auto loan data that our approach significantly speeds up learning compared with ignoring shared structure (i.e., using prior-independent Thompson sampling).

2. Problem Formulation

For ease of exposition, we primarily focus on a seller offering a *single* product at a time. Our approach and results generalize straightforwardly when multiple products are offered simultaneously, when a seller must also learn cross-product elasticities to capture substitution effects (see extension in Online Appendix F).

Notation: Throughout the paper, all vectors are column vectors by default. We define $[n]$ to be the set $\{1, 2, \ldots, n\}$ for any positive integer n. We use $\|x\|_u$ to denote the ℓ_u norm of a vector $x \in \mathbb{R}^d$, but we often omit the subscript when we refer to the ℓ_2 norm. For a matrix $X \in \mathbb{R}^{d\times d}$ $\|X\|_{op} := \max_{v\in\mathbb{R}^d:\|v\|=1}|v^\top Xv|$ is the operator norm of X. For a positive definite matrix $A \in \mathbb{R}^{d\times d}$ and vectors $x, y \in \mathbb{R}^d$, let $\|x\|_A$ denote the matrix norm $\sqrt{x^\top Ax}$ and $\langle x, y\rangle$ denote the inner product $x^\top y$. For two matrices A and B, we use $A \otimes B$ to denote their Kronecker product. We also denote $x \vee y$ and $x \wedge y$ as the maximum and minimum between $(x, y) \in \mathbb{R}$, respectively. We use the standard notation $O(\cdot), \Omega(\cdot)$, and $\Theta(\cdot)$ to characterize the asymptotic growth rate of a function (Cormen et al. 2009); when logarithmic factors are omitted, we use $\tilde{O}(\cdot), \tilde{\Omega}(\cdot)$, and $\tilde{\Theta}(\cdot)$. Finally, let $\lambda_{\min}(\cdot)$ and $\lambda_{\max}(\cdot)$ denote the minimum and maximum eigenvalues of a matrix, respectively.

2.1. Model

We first describe the classical dynamic pricing formulation for a single product; we then formalize our meta-learning formulation over a sequence of N products.

2.1.1. Classical Formulation. Consider a seller who offers a single product over a selling horizon of T periods. The seller can dynamically adjust the offered price in each period. At the beginning of each period $t \in [T]$, the seller observes a random feature vector (capturing exogenous and/or customer-specific features) that is independently and identically distributed from an unknown distribution. Upon observing the feature vector, the seller chooses a price for that period. The seller then observes the resulting demand, which is a noisy function of both the observed feature vector and the chosen price. The seller's revenue in each period is given by the chosen price multiplied by the corresponding realized demand. The goal in this setting is to develop a policy π that maximizes the seller's cumulative revenue by balancing exploration (learning the demand function) with exploitation (offering the estimated revenue-maximizing price).

2.1.2. Meta-Learning Formulation. We consider a seller who sequentially offers N related products, each with a selling horizon of T periods. For simplicity, a new product is not introduced until the life cycle of the previous product ends.[2] We call each product's life cycle an *epoch*; that is, there are N epochs that last T periods each. Each product (and corresponding epoch) is associated with a different (unknown) demand function and constitutes a different instance of the classical dynamic pricing problem described. We now formalize the problem.

In epoch $i \in [N]$ at time $t \in [T]$, the seller observes a random feature vector $x_{i,t} \in \mathbb{R}^d$, which is independently and identically distributed from a known distribution

$\mathcal{P}_i$. The seller then chooses a price $p_{i,t}$ for that period. Based on practical constraints, we assume that the allowable price range is bounded across periods and products, that is, $p_{i,t} \in [p_{\min}, p_{\max}]$ and $0 < p_{\min} < p_{\max} < \infty$. The seller then observes the resulting induced demand

$$D_{i,t}(p_{i,t}, x_{i,t}) = \langle \alpha_i, x_{i,t} \rangle + p_{i,t} \langle \beta_i, x_{i,t} \rangle + \varepsilon_{i,t},$$

where $\alpha_i \in \mathbb{R}^d$ and $\beta_i \in \mathbb{R}^d$ are unknown fixed constants throughout epoch i and $\varepsilon_{i,t} \sim \mathcal{N}(0, \sigma^2)$ is i.i.d. Gaussian noise with variance σ^2. This demand model was recently proposed by Ban and Keskin (2021) and captures several salient aspects. In particular, the observed feature vector $x_{i,t}$ in period t determines both the baseline demand (through the parameter α_i) and the price-elasticity of the demand (through the parameter β_i) of product i.

Example 1 (Rue La La). Rue La La sells a limited set of new products in multiday "events" (Ferreira et al. 2015). In this case, T is the number of price changes during the event (events are typically one to four days, and prices are updated no more than a few times a day), N is the number of events offered so far by the seller (note that $N \gg T$), and K is the number of simultaneously offered products in an event. For ease of exposition, we primarily consider $K = 1$, but Online Appendix F provides a straightforward extension to general values of K, accounting for substitution effects.

Remark 1 (Alternative Demand Models). Our demand model utilizes a continuous outcome variable, motivated by the setting in which many customers simultaneously view the same product with the same price in a given time unit. One can alternatively modify the demand model to follow a generalized linear model (e.g., logistic) to consider a binary purchase outcome variable for *each* customer. Our proposed algorithms easily generalize by appropriately modifying our Bayesian posterior update rules; however, we restrict our regret analysis to the linear case because ordinary least squares (OLS) Bayesian posterior updates have a closed form, yielding a tractable analysis.

2.1.3. Shared Structure. For ease of notation, we denote $\theta_i = (\alpha_i^\top \quad \beta_i^\top)^\top \in \mathbb{R}^{2d}$; following the classical formulation of dynamic pricing, θ_i is the unknown parameter vector that must be learned within a given epoch in order for the seller to maximize revenues over T periods. When there is no shared structure between the $\{\theta_i\}_{i=1}^N$, our problem reduces to N independent dynamic pricing problems.

However, we may expect that related products share a similar potential market and, thus, may have some shared structure that can be learned from previously offered products. We model this relationship by positing that the product demand parameter vectors $\{\theta_i\}_{i=1}^N$ are independent and identically distributed draws from a common unknown distribution, that is, $\theta_i \sim \mathcal{N}(\theta_*, \Sigma_*)$ for each $i \in [N]$.[3] As discussed earlier, knowledge of the distribution over the unknown demand parameters can inform the prior for Thompson sampling, thereby avoiding the need to use a conservative prior that can result in poor empirical performance (Honda and Takemura 2014, Liu and Li 2015). The mean of the shared distribution θ_* is unknown; we consider settings in which the covariance of this distribution Σ_* is known and unknown. We propose using meta-learning to learn this distribution from past epochs to inform and improve the current product's pricing strategy.

Remark 2 (Product Features). A complementary form of shared structure can be captured through product features. However, even after conditioning on observed product features, the demand functions for two products may behave very differently; for example, two black dresses may cater to very different types of customers or have very different price elasticities because of attributes such as fit or design that may be hard to capture as features. To capture product-specific (i.e., SKU-level) demand behaviors, we allow the *coefficients* of the demand function (e.g., price elasticity) to differ.

2.2. Assumptions

We now describe some mild assumptions on the parameters of the problem for our regret analysis.

Assumption 1 (Boundedness). *The support of the features are bounded, that is,*

$$\forall i \in [N], \ \forall t \in [T] \quad \| x_{i,t} \| \leq x_{\max}.$$

Furthermore, there exists a positive constant S such that $\|\theta_*\| \leq S$.

Our first assumption is that the observed feature vectors $\{x_{i,t}\}$ as well as the mean of the product demand parameters θ_* are bounded. This is a standard assumption made in the bandit and dynamic pricing literature, ensuring that the expected regret at any time step is bounded. This is likely satisfied because features and outcomes are typically bounded in practice.

Assumption 2 (Positive-Definite Feature Covariance). *The minimum eigenvalue of the feature covariance matrix $\mathbb{E}_{x_{i,t} \sim \mathcal{P}_i}[x_{i,t} x_{i,t}^\top]$ in every epoch $i \in [N]$ is lower bounded by some positive constant λ_0, that is,*

$$\min_{i \in [N]} \lambda_{\min}(\mathbb{E}_{x_{i,t} \sim \mathcal{P}_i}[x_{i,t} x_{i,t}^\top]) \ \geq \ \lambda_0.$$

Our second assumption imposes that the covariance matrix of the observed feature vectors $\mathbb{E}[x_{i,t} x_{i,t}^\top]$ in every epoch is positive-definite. This is a standard assumption for the convergence of OLS estimators; in

particular, our demand model is linear and, therefore, requires that no features are perfectly collinear in order to identify each product's true demand parameters.

Assumption 3 (Positive-Definite Prior Covariance). *The maximum and minimum eigenvalues of Σ_* are upper and lower bounded by positive constants $\bar{\lambda}$ and $\underline{\lambda}$, respectively, that is,*

$$\lambda_{\max}(\Sigma_*) \le \bar{\lambda}, \quad \lambda_{\min}(\Sigma_*) \ge \underline{\lambda}.$$

Our final assumption imposes that the covariance matrix of the random product demand parameter θ is also positive-definite and bounded. Again, this assumption ensures that each product's true demand parameter is identifiable using standard OLS estimators.

2.3. Background on Thompson Sampling with Known Prior

In this section, we consider the setting in which the true prior $\mathcal{N}(\theta_*, \Sigma_*)$ over the unknown product demand parameters is *known*. This setting informs our definition of the meta oracle and meta regret in the next section. When the prior is known, a natural candidate policy for minimizing Bayes regret is the Thompson sampling algorithm (Thompson 1933). The Thompson sampling algorithm adapted to our dynamic pricing setting for a single epoch $i \in [N]$ is formally given in Algorithm 1. Because the prior is known, there is no additional shared structure to exploit across products, so we can treat each epoch independently.

We denote $\mathrm{TS}(\mathcal{N}(\theta_*, \Sigma_*), \lambda_e)$, as the Thompson sampling algorithm with prior $\mathcal{N}(\theta_*, \Sigma_*)$ and a positive input parameter λ_e for initialization. In line with pricing algorithms in the literature (see, e.g., Keskin and Zeevi 2014, Ban and Keskin 2021), to ensure that we can obtain a well-defined OLS estimate of the underlying parameter at the end of an epoch, our algorithm initially performs random price exploration (alternating between $p_{\min}$ and $p_{\max}$) until the Fisher information matrix $V_{i,t} = \sum_{s=1}^{t} (x_{i,s}^\top \quad p_{i,s} x_{i,s}^\top)^\top (x_{i,s}^\top \quad p_{i,s} x_{i,s}^\top)$ has minimum eigenvalue of at least λ_e. Let $\mathcal{T}_i$ be the (random) length of this initialization period in epoch i,

$$\mathcal{T}_i = \arg\min_t \lambda_{\min}(V_{i,t}) \ge \lambda_e. \tag{1}$$

We show that $\mathcal{T}_i = \tilde{O}(1)$ with high probability (see Lemma 4 in Online Appendix A), and therefore, this initialization period forms a negligible portion of the epoch.

For each time step after initialization, $t \ge \mathcal{T}_i + 1$, the algorithm (1) samples the unknown product demand parameters $\mathring{\theta}_{i,t} = [\mathring{\alpha}_{i,t}; \mathring{\beta}_{i,t}]$ from the posterior $\mathcal{N}(\theta_{i,t}^{\mathrm{TS}}, \Sigma_{i,t}^{\mathrm{TS}})$ and (2) solves and offers the resulting optimal price based on the demand function given by the sampled parameters

$$p_{i,t}^{\mathrm{TS}} = \operatorname*{argmax}_{p \in [p_{\min}, p_{\max}]} \; p \cdot \langle \mathring{\alpha}_{i,t}, x_{i,t} \rangle + p^2 \cdot \langle \mathring{\beta}_{i,t}, x_{i,t} \rangle. \tag{2}$$

Upon observing the actual realized demand $D_{i,t}(p_{i,t}^{\mathrm{TS}}, x_{i,t})$, the algorithm computes the posterior $\mathcal{N}(\theta_{i,t+1}^{\mathrm{TS}}, \Sigma_{i,t+1}^{\mathrm{TS}})$ for round $t+1$. Specifically, using the update rule for Bayesian linear regression (Bishop 2006) and letting $m_{i,t}^{\mathrm{TS}} = (x_{i,t}^\top, p_{i,t}^{\mathrm{TS}} x_{i,t}^\top)^\top$, the posterior at time t is

$$\theta_{i,t}^{\mathrm{TS}} = \left(\Sigma_*^{-1} + \sigma \sum_{s=1}^{t-1} m_{i,s}^{\mathrm{TS}} (m_{i,s}^{\mathrm{TS}})^\top\right)^{-1} \left(\Sigma_*^{-1} \theta_* + \sigma \sum_{s=1}^{t-1} m_{i,s}^{\mathrm{TS}} D_{i,s}\right),$$

$$\Sigma_{i,t}^{\mathrm{TS}} = \left(\Sigma_*^{-1} + \sigma \sum_{s=1}^{t-1} m_{i,s}^{\mathrm{TS}} (m_{i,s}^{\mathrm{TS}})^\top\right)^{-1}.$$

The same algorithm is applied independently to each epoch $i \in [N]$.

Algorithm 1 ($\mathrm{TS}(\mathcal{N}(\theta_*, \Sigma_*), \lambda_e)$: Thompson Sampling Algorithm)

1. **Input:** The prior mean vector θ_* and covariance matrix Σ_*, the index i of epoch, the length of each epoch T, the noise parameter σ, exploration parameter λ_e.
2. **Initialization:** $t \leftarrow 1, (\theta_{i,t}^{\mathrm{TS}}, \Sigma_{i,t}^{\mathrm{TS}}) \leftarrow (\theta_*, \Sigma_*)$.
3. **while** $\lambda_{\min}\left(\sum_{s=1}^{t-1} (x_{i,s}^\top \quad p_{i,s} x^\top)^\top (x_{i,s}^\top \quad p_{i,s} x^\top)\right) < \lambda_e$ **do**
4. Observe feature vector $x_{i,t}$, and offer price $p_{i,t}^{\mathrm{TS}} \leftarrow \begin{cases} p_{\max} & \text{if } t \text{ is even,} \\ p_{\min} & \text{otherwise.} \end{cases}$
5. Observe demand $D_{i,t}(p_{i,t}^{\mathrm{TS}}, x_{i,t})$, and compute the posterior $\mathcal{N}(\theta_{i,t+1}^{\mathrm{TS}}, \Sigma_{i,t+1}^{\mathrm{TS}})$.
6. $t \leftarrow t+1$
7. **end while**
8. **while** $t \le T$ **do**
9. Observe feature vector $x_{i,t}$.
10. Sample parameter $\mathring{\theta}_{i,t} \leftarrow [\mathring{\alpha}_{i,t}; \mathring{\beta}_{i,t}] \sim \mathcal{N}(\theta_{i,t}^{\mathrm{TS}}, \Sigma_{i,t}^{\mathrm{TS}})$.
11. $p_{i,t}^{\mathrm{TS}} \leftarrow \arg\max_{p \in [p_{\min}, p_{\max}]} \; p \cdot \langle \mathring{\alpha}_{i,t}, x_{i,t} \rangle + p^2 \cdot \langle \mathring{\beta}_{i,t}, x_{i,t} \rangle$.
12. Observe demand $D_{i,t}(p_{i,t}^{\mathrm{TS}}, x_i)$, and compute the posterior $\mathcal{N}(\theta_{i,t+1}^{\mathrm{TS}}, \Sigma_{i,t+1}^{\mathrm{TS}})$.
13. $t \leftarrow t+1$
14. **end while.**

As evidenced by the large literature on the practical success of Thompson sampling (Chapelle and Li 2011, Russo and Van Roy 2014, Ferreira et al. 2018), Algorithm 1 is a very attractive choice for implementation in practice.

Algorithm 1 attains a strong performance guarantee under the classical formulation compared with an *oracle* that knows all N product demand parameters $\{\theta_i\}_{i=1}^N$ in advance. In particular, the oracle would offer the expected optimal price in each period $t \in [T]$ in epoch $i \in [N]$, that is,

$$\begin{aligned} p_{i,t}^* &= \arg\max_{p \in [p_{\min}, p_{\max}]} \; p \cdot \mathbb{E}_\varepsilon[D_{i,t}(p, x_{i,t})] \\ &= \arg\max_{p \in [p_{\min}, p_{\max}]} \; p \langle \alpha_i, x_{i,t} \rangle + p^2 \langle \beta_i, x_{i,t} \rangle. \end{aligned} \tag{3}$$

The resulting *Bayes regret* (Russo and Van Roy 2014) of a policy π relative to the oracle is

$$\text{Bayes Regret}_{N,T}(\pi) = \mathbb{E}_{\theta,x,\varepsilon}\left[\sum_{i=1}^{N}\sum_{t=1}^{T} p^*_{i,t} D(p^*_{i,t}, x_{i,t}) - \sum_{i=1}^{N}\sum_{t=1}^{T} p^{\pi}_{i,t} D(p^{\pi}_{i,t}, x_{i,t})\right], \quad (4)$$

where the expectation is taken with respect to the unknown product demand parameters, the observed random feature vectors, and the noise in the realized demand. The following theorem bounds the Bayes regret of the Thompson sampling dynamic pricing algorithm:

Theorem 1. *When the prior over the demand parameters is known, Algorithm 1 satisfies*

$$\text{Bayes Regret}_{N,T}(\pi) = \tilde{O}\left(d^{\frac{3}{2}} N\sqrt{T}\right),$$

Theorem 1 follows from a similar argument used for the linear bandit setting presented in Russo and Van Roy (2014), coupled with standard concentration bounds for multivariate normal distributions. The proof is given in Online Appendix A for completeness. Note that the regret scales linearly in N because each epoch is an independent learning problem.

Remark 3. Prior-independent Thompson sampling (Agrawal and Goyal 2013) achieves a Bayes regret of $\tilde{O}(d^2 N\sqrt{T})$, which is comparable to the performance of Algorithm 1. However, we document a substantial gap in empirical performance between the two approaches in Section 5, motivating our study of learning the prior.

2.4. Meta Oracle and Meta Regret

We cannot directly implement Algorithm 1 in our setting because the prior over the product demand parameters $\mathcal{N}(\theta_*, \Sigma_*)$ is unknown. In this paper, we seek to learn the prior (shared structure) across products in order to leverage the superior performance of Thompson sampling with a known prior. Thus, this is a natural question to ask: what is the price of not knowing the prior in advance?

To answer this question, we first define our performance metric. Because our goal is to converge to the policy given in Algorithm 1 (which knows the true prior), we define this policy as our *meta oracle*.[4] Comparing the revenue of our policy relative to the meta oracle leads naturally to the definition of *meta regret* $\mathcal{R}_{N,T}$ for a policy π, that is,

$$\mathcal{R}_{N,T}(\pi) = \mathbb{E}_{\theta,x,\varepsilon}\left[\sum_{i=1}^{N}\sum_{t=1}^{T} p^{\text{TS}}_{i,t} D(p^{\text{TS}}_{i,t}, x_{i,t}) - \sum_{i=1}^{N}\sum_{t=1}^{T} p^{\pi}_{i,t} D(p^{\pi}_{i,t}, x_{i,t})\right],$$

where the expectation is taken with respect to the unknown product demand parameters, the observed random feature vectors, and the noise in the realized demand.

Note that prior-independent Thompson sampling and Upper Confidence Bound (UCB) treat each epoch independently and would, thus, achieve meta regret that grows linearly in N. Our goal is to design a policy with meta regret that grows sublinearly in N. Recall that Theorem 1 bounds the Bayes regret of Thompson sampling with a known prior as $\tilde{O}(N\sqrt{T})$. Thus, if our meta regret (i.e., the performance of our meta-learning policy relative to Algorithm 1) grows sublinearly in N, then the price of not knowing the prior $\mathcal{N}(\theta_*, \Sigma_*)$ in advance is negligible in experiment-rich environments (large N) compared with the cost of learning the demand parameter for each product (the Bayes regret of Algorithm 1).

The values of the prior mean θ_* as well as the actual product demand parameter vectors $\{\theta_i\}_{i=1}^N$ are unknown; we consider two settings: known and unknown Σ_* (covariance of the prior).

Remark 4 (Choice of Meta Oracle). To the best of our knowledge, the optimal prior to use for Thompson sampling remains a difficult, open problem. Existing theory shows (in limited settings) that priors that fail to place sufficient mass on the true parameter fare poorly: the closest setting to ours is the linear bandit construction in proposition 3.1 of Hamidi and Bayati (2020), which shows that prior-dependent Thompson sampling with a misspecified prior can achieve regret that scales exponentially in d; theorem 1 of Liu and Li (2015) and theorem 2 of Honda and Takemura (2014) also provide illustrative constructions with the same insight. In the other extreme, many empirical evaluations suggest that overly conservative priors (such as prior-independent approaches) also fare poorly relative to using the true prior (see, e.g., section 6 of Bastani et al. (2021), the discussions in Chapelle and Li (2011), or our numerical results in Section 5). As a result, we choose Thompson sampling with the true prior as our meta oracle. However, one can choose alternative meta oracles—for example, one that "widens" the true prior to place more weight on parameters that may induce higher regret—and implementing such a meta oracle would still likely require learning the true prior, which is our primary contribution.

2.4.1. Nonanticipating Policies. We restrict ourselves to the family of nonanticipating policies $\Pi : \pi = \{\pi_{i,t}\}$ that form a sequence of random functions $\pi_{i,t}$ that depend only on price and demand observations collected until time t in epoch i (including all times $t \in [T]$ from prior epochs), and feature vector observations up to time $t+1$ in epoch i. In particular, let $\mathcal{H}_{0,0} = (x_{1,1})$ and

$\mathcal{H}_{i,t} = (p_{1,1}, p_{1,2}, \cdots, p_{i,t}, D_{1,1}, D_{1,2}, \cdots, D_{i,t}, x_{1,1}, x_{1,2}, \cdots, x_{i,t+1})$ denote the history of prices and corresponding demand realizations from prior epochs and time periods as well as the observed feature vectors up to the next time period; let $\mathcal{F}_{i,t}$ denote the σ-field generated by $\mathcal{H}_{i,t}$. Then, we impose that $\pi_{i,t+1}$ is $\mathcal{F}_{i,t}$ measurable.

3. Meta-DP Algorithm

We begin with the case in which the prior's covariance matrix Σ_* is known and describe the meta dynamic pricing (Meta-DP) algorithm for this setting. We consider the case of unknown Σ_* in the next section.

3.1. Overview

The Meta-DP algorithm begins by using initial product epochs as an exploration phase to initialize our estimate of the prior mean θ_*. These exploration epochs use the prior-independent Thompson sampling algorithm to ensure no more than $\tilde{O}(d^2\sqrt{T})$ meta regret for each epoch. After this initial exploration period, our algorithm sequentially updates the estimated prior and leverages this estimate in each subsequent epoch. The key technical challenge is that the estimated prior has finite-sample estimation error, resulting in a Thompson sampling instance with a misspecified prior. We introduce a prior alignment proof technique to show that, despite prior misspecification, our Meta-DP algorithm still achieves meta regret that grows sublinearly in N.

3.2. Algorithm

The Meta-DP algorithm is presented in Algorithm 2. We first define some additional notation and then describe the algorithm in detail.

3.2.1. Additional Notation. Throughout the rest of the paper, we use $m_{i,t} = (x_{i,t}^\top \quad p_{i,t}x_{i,t}^\top)^\top$ to denote the price and feature information and $V_{i,t} = \sum_{\tau=1}^{t} m_{i,t}m_{i,t}^\top$ to denote the Fisher information matrix of round t in epoch i for all $i \in [N]$ and $t \in [T]$.

3.2.2. Algorithm Description. The first N_0 epochs are treated as exploration epochs, in which we define

$$N_0 = 4c_2^2 d\mathcal{T}_e^2 \log_e(4dN^2T)\log_e(2NT) = \tilde{O}(d), \tag{5}$$

where $\mathcal{T}_e = \max\{6\log_{e/2}(dNT)/c_1, 2\lambda_e/c_0\} = \tilde{O}(1)$ ($\mathcal{T}_e$ is a high probability upper bound on all $\mathcal{T}_i$ s, see Lemma 4 in Online Appendix A), and the constant is given by

$$c_2 = \frac{32\sqrt{x_{\max}^2(1+p_{\max}^2)(\sigma^2\lambda_e^{-1} + 5\overline{\lambda})}}{\lambda_e \underline{\lambda}\sigma^2}.$$

As described in the overview, the Meta-DP algorithm proceeds in two phases. In particular, we distinguish the following two cases for each epoch i:

1. Epoch $\mathbf{i} \le \mathbf{N}_0$: the Meta-DP algorithm runs the prior-independent Thompson sampling algorithm (Agrawal and Goyal 2013, Abeille and Lazaric 2017) TS($\mathcal{N}(0, \Psi I_{2d}), \lambda_e$), where

$$\Psi = p_{\max}\sigma\sqrt{2d\log_e(T(1+x_{\max}^2 p_{\max}^2(1+p_{\max}^2)T))} + \sqrt{20\overline{\lambda}d\log_e(2T)}.$$

This is simply Algorithm 1 with a conservative prior (variance is a function of the horizon T).

2. Epoch $\mathbf{i} > \mathbf{N}_0$: the Meta-DP algorithm first computes the OLS estimate of the true parameter for each previous epoch $j < i$. It then averages these parameter estimates to form an estimator $\hat{\theta}_i$ of the prior mean θ_*, that is,

$$\hat{\theta}_i = \frac{\sum_{j=1}^{i-1} V_{j,T}^{-1}\left(\sum_{t=1}^{T} D_{j,t}(p_{j,t}, x_{j,t})m_{j,t}\right)}{i-1}. \tag{6}$$

Then, the Meta-DP algorithm runs Thompson sampling (Algorithm 1) with the estimated prior $\mathcal{N}(\hat{\theta}_i, \Sigma_*)$, that is, TS($\mathcal{N}(\hat{\theta}_i, \Sigma_*), \lambda_e$). Specifically, after some random initialization steps (these steps are identical to our meta oracle), our Meta-DP algorithm (1) samples the unknown product demand parameters $\mathring{\theta}_{i,t} = [\mathring{\alpha}_{i,t}; \mathring{\beta}_{i,t}]$ from its posterior $\mathcal{N}(\theta_{i,t}^{\mathrm{MD}}, \Sigma_{i,t}^{\mathrm{MD}})$ and (2) solves and offers the resulting optimal price based on the demand function given by the sampled parameters

$$p_{i,t} = \underset{p\in[p_{\min}, p_{\max}]}{\arg\max}\ p\cdot\langle\mathring{\alpha}_{i,t}, x_{i,t}\rangle + p^2\cdot\langle\mathring{\beta}_{i,t}, x_{i,t}\rangle. \tag{7}$$

Upon observing the actual realized demand $D_{i,t}(p_{i,t}, x_{i,t})$, the algorithm computes the posterior $\mathcal{N}(\theta_{i,t+1}^{\mathrm{MD}}, \Sigma_{i,t+1}^{\mathrm{MD}})$ for round $t+1$.

Algorithm 2 (Meta-Dynamic Pricing Algorithm)

1. **Input:** The prior covariance matrix Σ_*, the total number of epochs N, the length of each epoch T, the noise parameter σ, and the set of feasible prices $[p_{\min}, p_{\max}]$.
2. **Initialization:** N_0 as defined in Eq. (5).
3. **for** each epoch $i = 1, \ldots, N$ **do**
4. **if** $i \le N_0$ **then**
5. Run TS$\left(\mathcal{N}(0, \Psi), \lambda_e\right)$.
6. **else**
7. Update $\hat{\theta}_i$ according to Equation (6), and run TS$\left(\mathcal{N}(\hat{\theta}_i, \Sigma_*), \lambda_e\right)$.
8. **end if**
9. **end for**

We now state our main result upper bounding the meta regret of our Meta-DP algorithm (Algorithm 2). The proof is provided in Section 3.3 and Online Appendix C.

Theorem 2. *The meta regret of the proposed Meta-DP algorithm satisfies*

$$\mathcal{R}_{N,T}(\texttt{Meta-DP}\ \textit{algorithm}) = \begin{cases} \tilde{O}(d^2 N\sqrt{T}) & \text{when } N < N_0 \\ \tilde{O}(d^2\sqrt{NT}) & \text{otherwise} \end{cases} = \tilde{O}(d^2\sqrt{NT} + d^3\sqrt{T}).$$

It is worthwhile to compare the bound in Theorem 2 to the $\tilde{O}(d^2 N\sqrt{T})$ meta regret bound for prior-independent Thompson sampling (Lemma 11 in Online Appendix C). When $N \lesssim \tilde{O}(d)$, our bound matches that of prior-independent Thompson sampling because we simply treat all our epochs as exploration epochs. In the large N regime, our meta regret scales as $\tilde{O}(d^2\sqrt{NT})$. Thus, our approach of learning the prior is particularly valuable in experiment-rich settings ($N \gg d$). Combining the two regimes yields a bound that is sublinear in both N and T.

Theorem 2 is somewhat surprising in the context of a growing theoretical literature that suggests that a misspecified prior can result in very poor regret for prior-dependent Thompson sampling (see, e.g., Honda and Takemura 2014, Liu and Li 2015, Hamidi and Bayati 2020). Indeed, one may expect that the misspecification induced by using the prior $\mathcal{N}(\hat{\theta}_i, \Sigma_*)$ instead of $\mathcal{N}(\theta_*, \Sigma_*)$ can be substantial because the ratio between these two probability density functions is unbounded when $\hat{\theta}_i \neq \theta_*$. Yet, using our prior alignment proof strategy (described in the next section), we establish that Thompson sampling is remarkably robust to misspecification of the prior *mean*, lending theoretical support to previous empirical observations (Bastani et al. 2021).

3.3. Prior Alignment Proof Strategy

Because we only have a logarithmic number (in N and T) of exploration epochs, the meta regret accrued from these epochs is $\tilde{O}(d^2 N_0\sqrt{T})$ (see Lemma 11 in Online Appendix C).

In each nonexploration epoch $i > N_0$, the meta oracle starts with the true prior $\mathcal{N}(\theta_*, \Sigma_*)$ although our algorithm $\texttt{Meta-DP}$ starts with the estimated prior $\mathcal{N}(\hat{\theta}_i, \Sigma_*)$. The following lemma (whose proof is in Online Appendix B) bounds the error of the estimated prior mean with high probability:

Lemma 1. *For any fixed $i \geq 2$ and $\delta \in [0, 2/e]$, with probability at least $1 - \delta - 2/(N^2T^2)$,*

$$\| \hat{\theta}_i - \theta_* \| \leq 8\sqrt{\frac{2(\sigma^2/\lambda_e + 5\bar{\lambda})d\log_e(4d/\delta)}{i}}.$$

Thus, the key challenge in proving Theorem 2 is bounding the difference in regret incurred by using a Thompson sampling algorithm with a boundedly misspecified prior. We introduce a new prior alignment proof technique to address this challenge. At a high level, we show that, after the $\mathcal{T}_i$ exploration time steps, the distributions of the meta oracle's (random) posterior estimate $\theta^{\text{TS}}_{i,\mathcal{T}_i+1}$ and $\texttt{Meta-DP}$'s (random) posterior estimate $\theta^{\text{MD}}_{i,\mathcal{T}_i+1}$ are close. More specifically, there is a continuum of realizations of the stochastic noise (in the observed demands) such that $\texttt{Meta-DP}$ achieves the *same* posterior estimate $\theta^{\text{MD}}_{i,\mathcal{T}_i+1} = \theta^{\text{TS}}_{i,\mathcal{T}_i+1}$ despite starting with a different prior; when such a match occurs, the expected regret moving forward from time $\mathcal{T}_i + 1, \cdots, T$ is the same for both policies. Using this approach, the regret of our $\texttt{Meta-DP}$ algorithm can be expressed as a weighted distribution of the regret of the meta oracle (which we bounded in Theorem 1).

More specifically, the following lemma (whose proof is in Online Appendix C) establishes the difference in Bayesian posteriors between the meta oracle and our Meta-DP algorithm. Note that only the means of the posterior differ but the variance is the same.

Lemma 2. *Conditioned on θ_i and $x_{i,1}, \ldots, x_{i,\mathcal{T}_i}$, the posteriors of the meta oracle and our Meta-DP algorithm satisfy*

$$\theta^{\text{TS}}_{i,\mathcal{T}_i+1} - \theta^{\text{MD}}_{i,\mathcal{T}_i+1} = \left(\Sigma_*^{-1} + \sigma\sum_{t=1}^{\mathcal{T}_i} m_{i,t}m_{i,t}^\top\right)^{-1} \left(\Sigma_*^{-1}(\theta_* - \hat{\theta}_i) + \sigma\sum_{t=1}^{\mathcal{T}_i} m_{i,t}\left(\varepsilon^{\text{TS}}_{i,t} - \varepsilon^{\text{MD}}_{i,t}\right)\right),$$

$$\Sigma^{\text{TS}}_{i,\mathcal{T}_i+1} = \Sigma^{\text{MD}}_{i,\mathcal{T}_i+1}.$$

Now, consider any nonexploration epoch $i \geq N_0 + 1$. If, upon completion of all exploration steps at time $\mathcal{T}_i + 1$, we have that the posteriors of the meta oracle and our Meta-DP algorithm coincide—that is, $(\theta^{\text{MD}}_{i,\mathcal{T}_i+1}, \Sigma^{\text{MD}}_{i,\mathcal{T}_i+1}) = (\theta^{\text{TS}}_{i,\mathcal{T}_i+1}, \Sigma^{\text{TS}}_{i,\mathcal{T}_i+1})$—then both policies would achieve the same expected revenue over the time periods $\mathcal{T}_i + 1, \cdots, T$. By Lemma 2, we know that $\Sigma^{\text{TS}}_{i,\mathcal{T}_i+1} = \Sigma^{\text{MD}}_{i,\mathcal{T}_i+1}$ always, so all that remains is establishing when $\theta^{\text{TS}}_{i,\mathcal{T}_i+1} = \theta^{\text{MD}}_{i,\mathcal{T}_i+1}$.

Because the two algorithms begin with different priors but encounter the same covariates $\{x_{i,t}\}_{t=1}^T$ and take the same decisions in $t \in \{1, \cdots, \mathcal{T}_i\}$, their posteriors can only align at time $\mathcal{T}_i + 1$ because of the stochasticity in the observations $\varepsilon_{i,t}$. For convenience, denote the noise terms from $t \in \{1, \cdots, \mathcal{T}_i\}$ of the meta oracle and the Meta-DP algorithm respectively as

$$\chi_i^{\text{TS}} = (\varepsilon^{\text{TS}}_{i,1} \ldots \varepsilon^{\text{TS}}_{i,\mathcal{T}_i})^\top, \tag{8}$$

$$\chi_i^{\text{MD}} = (\varepsilon^{\text{MD}}_{i,1} \ldots \varepsilon^{\text{MD}}_{i,\mathcal{T}_i})^\top. \tag{9}$$

Furthermore, let $M_i = (m_{i,1} \dots m_{i,\mathcal{T}_i}) \in \mathbb{R}^{2d \times \mathcal{T}_i}$. Lemma 2 indicates that, if

$$\chi_i^{\mathrm{MD}} - \chi_i^{\mathrm{TS}} = \frac{1}{\sigma}(M_i^\top M_i)^{-1} M_i^\top \Sigma_*^{-1}(\theta_* - \hat{\theta}_i), \tag{10}$$

then the posteriors of both algorithms align with $\theta_{i,\mathcal{T}_i+1}^{\mathrm{TS}} = \theta_{i,\mathcal{T}_i+1}^{\mathrm{MD}}$. Thus, for every realization of the meta oracle's noise terms χ_i^{TS} and the prior mean estimation error $\theta_* - \hat{\theta}_i$, there exists a well-defined and feasible choice of Meta-DP algorithm's error χ_i^{MD} that allows the two posteriors to coincide. Furthermore, by Lemma 1, $\|\theta_* - \hat{\theta}_i\|$ is bounded as a function of $\sqrt{1/i}$ with high probability, ensuring that the difference in noise terms $\chi_i^{\mathrm{MD}} - \chi_i^{\mathrm{TS}}$ needed to achieve alignment is small for later epochs (as i grows large). With this observation, we can perform a change of measure over our noise terms and integrate over the resulting distributions, yielding the desired bound on the meta regret. The proof is provided in Online Appendix C.

Remark 5. Our prior alignment approach may be of general interest for analyzing the regret of misspecified Thompson sampling instances. Russo and Van Roy (2014) propose a related but different approach in Section 3.1 of their paper. Specifically, they relate the regret of implementing $TS(\mathcal{N}(\hat{\theta}_i, \Sigma_*), \lambda_e)$ in an environment with true prior $\mathcal{N}(\theta_*, \Sigma_*)$ to the regret of $TS(\mathcal{N}(\hat{\theta}_i, \Sigma_*), \lambda_e)$ in an environment with a different true prior $\mathcal{N}(\hat{\theta}_i, \Sigma_*)$. In contrast, we wish to compare the regret of implementing $TS(\mathcal{N}(\hat{\theta}_i, \Sigma_*), \lambda_e)$ (Meta-DP, Algorithm 2) and $TS(\mathcal{N}(\theta_*, \Sigma_*), \lambda_e)$ (meta oracle, Algorithm 1) in the same environment with true prior $\mathcal{N}(\theta_*, \Sigma_*)$. We cannot adopt their approach because one must additionally quantify the difference in regret between TS algorithms learning in environments with different true priors; although this regret difference clearly scales sublinearly in T, we require a bound that limits to zero as the difference in priors $\|\hat{\theta}_i - \theta_*\| \to 0$ (as $i \to \infty$). This requirement is because even a constant nonzero difference in regret between the meta oracle and our Meta-DP algorithm would result in $O(N)$ meta regret over N epochs. To our knowledge, it is an open problem to derive such a bound. Our prior alignment sidesteps this issue by directly relating $TS(\mathcal{N}(\hat{\theta}_i, \Sigma_*), \lambda_e)$ and $TS(\mathcal{N}(\theta_*, \Sigma_*), \lambda_e)$ in an environment with true prior $\mathcal{N}(\theta_*, \Sigma_*)$.

4. Meta-DP++ Algorithm

In this section, we consider the setting in which the prior covariance matrix Σ_* is also unknown. We propose the Meta-DP++ algorithm, which builds on top of the Meta-DP algorithm and additionally estimates the unknown prior covariance Σ_*.

4.1. Overview

The Meta-DP++ algorithm also begins by using initial product epochs as an exploration phase to initialize our estimate of the prior mean θ_* and covariance Σ_*. After this initial exploration period, our algorithm sequentially updates the estimated prior and leverages this estimate in each subsequent epoch. Once again, the estimated prior has a finite-sample estimation error, resulting in a Thompson sampling instance with a misspecified prior. The key challenge compared with the previous section is that we can no longer exactly "align" our algorithm's posterior with that of the meta oracle when Σ_* is also estimated. We leverage importance sampling arguments from off-policy evaluation to bound the additional meta regret accrued because of this mismatch. Importantly, to ensure that our importance weights remain well behaved, we *widen* the estimated covariance via a correction term that scales as the finite-sample estimation error of estimating $\hat{\Sigma}_*$.

4.2. Algorithm

The Meta-DP++ algorithm is presented in Algorithm 3. We first define some additional notation and then describe the algorithm in detail.

4.2.1. Additional Notation. As with the Meta-DP algorithm, at the beginning of each epoch $i \in [N]$, we update our estimate $\hat{\theta}_i$ of the prior mean θ_* according to Equation (6). To estimate Σ_*, we need unbiased and *independent* estimates for the unknown true demand parameter realizations θ_i across epochs.[5] We use the initialization steps $t \in [\mathcal{T}_i]$ to produce an estimate $\dot{\theta}_i$ for θ_i, that is,

$$\dot{\theta}_i = V_{i,\mathcal{T}_i}^{-1}\Big(\sum_{t=1}^{\mathcal{T}_i} D_{i,t}(p_{i,t}, x_{i,t}) m_{i,t}\Big).$$

4.2.2. Algorithm Description. The first N_1 epochs are treated as exploration epochs, in which we employ the prior-independent Thompson sampling algorithm. We define

$$\begin{aligned} N_1 = \max\{&N_0,\ 256 c_3^2 d^3 \mathcal{T}_e^2 \log_e^3(4dN^2T), \\ &c_4^2 d^4 T^2 \log_e^3(2N^2T)\} = \tilde{O}(d^4T^2), \end{aligned} \tag{11}$$

and the constants are given by

$$\begin{aligned} c_3 &= \frac{16\sqrt{\sigma^2\lambda_e^{-1} + 5\overline{\lambda}}}{\sigma\lambda_e\underline{\lambda}} + \frac{256(\overline{\lambda}\lambda_e^2 + 16\sigma^2)}{\lambda_e^2\underline{\lambda}^2} \\ &\quad \left(\frac{8p_{\max}x_{\max}\sqrt{(1+p_{\max}^2)}}{\lambda_e} + \frac{S}{\sigma\lambda_e}\right), \\ c_4 &= \frac{10^4\sigma(\overline{\lambda}\lambda_e^2 + 16\sigma^2)}{\lambda_e^2\underline{\lambda}^2}. \end{aligned}$$

Note that we now require $\tilde{O}(\min\{N, d^4T^2\})$ exploration epochs, whereas we only required $\tilde{O}(d^2)$ exploration epochs for the Meta-DP algorithm.

As described in the overview, the Meta-DP++ algorithm proceeds in two phases:

1. Epoch $\mathbf{i} \le \mathbf{N}_1$: the Meta-DP++ algorithm runs the prior-independent Thompson sampling algorithm (Agrawal and Goyal 2013, Abeille and Lazaric 2017) $\mathrm{TS}(\mathcal{N}(0, \Psi I_{2d}), \lambda_e)$, where

$$\Psi = p_{\max}\sigma\sqrt{2d\log_e(T(1 + x_{\max}^2 p_{\max}^2(1 + p_{\max}^2)T))} + \sqrt{20\overline{\lambda}d\log_e(2T)}.$$

This is simply Algorithm 1 with a conservative prior (variance is a function of the horizon T).

2. Epoch $\mathbf{i} > \mathbf{N}_1$: the Meta-DP++ algorithm computes an estimator $\hat{\theta}_i$ of the prior mean θ_* using Equation (6) (same as the Meta-DP algorithm) and an estimator $\hat{\Sigma}_i$ of the prior covariance Σ_* as

$$\hat{\Sigma}_i = \frac{1}{i-2}\sum_{j=1}^{i-1}\left(\dot{\theta}_j - \frac{\sum_{k=1}^{i-1}\dot{\theta}_k}{i-1}\right)\left(\dot{\theta}_j - \frac{\sum_{k=1}^{i-1}\dot{\theta}_k}{i-1}\right)^{\top} - \frac{\sigma^2\sum_{j=1}^{i-1}\mathbb{E}[V_{j,T_j}^{-1}]}{i-1}. \tag{12}$$

The second term $\sigma^2\sum_{j=1}^{i-1}\mathbb{E}[V_{j,T_j}^{-1}]/(i-1)$ accounts for the estimation error in $\{\dot{\theta}_j\}_{j=1}^{i-1}$.

As noted earlier, we then widen our estimator to account for a finite-sample estimation error:

$$\hat{\Sigma}_i^w = \hat{\Sigma}_i + \frac{128(\overline{\lambda}\lambda_e^2 + 16\sigma^2 d)}{\lambda_e^2}\sqrt{\frac{5d\log_e(2N^2T)}{i}}\cdot I_{2d}, \tag{13}$$

where I_{2d} is the $(2d)$-dimensional identity matrix.

Then, the Meta-DP++ algorithm runs Thompson sampling (Algorithm 1) with the estimated prior $\mathcal{N}(\hat{\theta}_i, \hat{\Sigma}_i^w)$, that is, $\mathrm{TS}(\mathcal{N}(\hat{\theta}_i, \hat{\Sigma}_i^w), \lambda_e)$. Specifically, after some random initialization steps (these steps are identical to our meta oracle), our Meta-DP++ algorithm (1) samples the unknown product demand parameters $\mathring{\theta}_{i,t} = [\mathring{\alpha}_{i,t}; \mathring{\beta}_{i,t}]$ from the posterior $\mathcal{N}(\theta_{i,t}^{\mathrm{MDP}}, \Sigma_{i,t}^{\mathrm{MDP}})$ and (2) solves and offers the resulting optimal price based on the demand function given by the sampled parameters

$$p_{i,t} = \underset{p\in[p_{\min}, p_{\max}]}{\arg\max}\ p\cdot\langle\mathring{\alpha}_{i,t}, x_{i,t}\rangle + p^2\cdot\langle\mathring{\beta}_{i,t}, x_{i,t}\rangle. \tag{14}$$

Upon observing the actual realized demand $D_{i,t}(p_{i,t}, x_{i,t})$, the algorithm computes the posterior $\mathcal{N}(\theta_{i,t+1}^{\mathrm{MDP}}, \Sigma_{i,t+1}^{\mathrm{MDP}})$ for round $t+1$.

Algorithm 3 (Meta-Dynamic Pricing Algorithm)

1. **Input:** The total number of products N, the length of each epoch T, the noise parameter σ, and the set of feasible prices $[p_{\min}, p_{\max}]$.
2. **for** epoch $i = 1, \ldots, N$ **do**
3. **if** $i \le N_1$ **then**
4. Run $\mathrm{TS}\left(\mathcal{N}(0, \Psi), \lambda_e\right)$.
5. **else**
6. Update $\hat{\theta}_i$ and $\hat{\Sigma}_i$ according to Equations (6) and (12), respectively.
7. Compute widened prior mean estimate $\hat{\Sigma}_i^w$ according to Equation (13).
8. Run $\mathrm{TS}\left(\mathcal{N}(\hat{\theta}_i, \hat{\Sigma}_i^w), \lambda_e\right)$.
9. **end if**
10. **end for**

We now state our main result upper bounding the meta regret of our Meta-DP++ algorithm (Algorithm 3). The proof is provided in Section 4.3 and Online Appendix E.

Theorem 3. *The meta regret of the proposed Meta-DP++ algorithm satisfies*

$$\mathcal{R}_{N,T}(\texttt{Meta-DP++}\ \textit{algorithm}) = \tilde{O}\left(\min\{d^2NT^{\frac{1}{2}},\ d^4N^{\frac{1}{2}}T^{\frac{3}{2}}\}\right) = \tilde{O}\left(d^3(NT)^{\frac{5}{6}}\right).$$

It is worthwhile to compare the bound in Theorem 3 to the $\tilde{O}(d^2N\sqrt{T})$ meta regret bound for prior-independent Thompson sampling (Lemma 11 in Online Appendix C). When $N \lesssim \tilde{O}(d^4T^2)$, our bound matches that of prior-independent Thompson sampling because we simply treat all our epochs as exploration epochs. In the large N regime, our meta regret scales as $\tilde{O}(d^4N^{\frac{1}{2}}T^{\frac{3}{2}})$. Thus, our approach of learning the prior is particularly valuable in settings with many short-horizon experiments ($N \gg T$). For instance, as discussed in Example 1, sellers like Rue La La host many events, offering new items with short selling seasons. Combining the two regimes yields a bound that is sublinear in both N and T.

4.3. Proof Strategy

The number of exploration epochs N_1 is logarithmic in N but quadratic in T. This motivates the analysis of two cases: (i) when the number of epochs $N < N_1 = \tilde{O}(d^4T^2)$, the meta regret guarantees given by existing prior-independent approaches are already good; (ii) when we transition to an experiment-rich environment with $N > N_1$, the meta regret accrued from these epochs is small because their cardinality scales logarithmically in N (see argument in Online Appendix E). We now focus on the latter case in which N is large.

Once again, following the proof strategy employed for the Meta-DP algorithm, we employ prior alignment to match the means of the meta oracle's (random) posterior estimate and Meta-DP++'s (random) posterior estimates. However, because Σ_* is known in the previous section, matching the posterior means $\theta^{\mathrm{MD}}_{i,\mathcal{T}_i+1} = \theta^{\mathrm{TS}}_{i,\mathcal{T}_i+1}$ implies equality of the *entire distribution* of the posterior (see Lemma 2). This equivalence allows us to exactly equate the expected regret (after alignment) for the meta oracle and our Meta-DP algorithm.

However, when Σ_* is unknown, matching the posterior means $\theta^{\mathrm{MDP}}_{i,\mathcal{T}_i+1} = \theta^{\mathrm{TS}}_{i,\mathcal{T}_i+1}$ no longer implies that the posterior distributions are equal. Furthermore, because the Bayesian update for the covariance matrix does not depend on the noise terms (it depends only on the observed covariates and chosen prices), we cannot use any alignment strategy based on χ^{TS}_i and χ^{MDP}_i to get exact equivalence of the posterior distributions. Thus, the key added challenge in proving Theorem 3 is bounding the difference in regret between our Meta-DP++ algorithm and the meta oracle *after* alignment of the means of their posteriors at time $t = \mathcal{T}_i$.

Specifically, in each nonexploration epoch $i > N_1$, the meta oracle starts with the true prior $\mathcal{N}(\theta_*, \Sigma_*)$ although our algorithm Meta-DP++ starts with the (widened) estimated prior $\mathcal{N}(\hat{\theta}_i, \hat{\Sigma}^w_i)$. Lemma 1 from the previous section already provides a bound on $\|\hat{\theta}_i - \theta_*\|$, and the following lemma (whose proof is in Online Appendix D) bounds the error of the estimated covariance $\|\hat{\Sigma}_i - \Sigma_*\|$ (and, thus, the error of our widened covariance $\|\hat{\Sigma}^w_i - \Sigma_*\|$) with high probability:

Lemma 3. *For any fixed $i \geq 3$ and $\delta \in [0, 2/e]$, with probability at least $1 - 2\delta - 2/(N^2T^2)$,*

$$\| \hat{\Sigma}_i - \Sigma_* \|_{op} \leq \frac{128(\overline{\lambda}\lambda_e^2 + 16\sigma^2 d)}{\lambda_e^2} \left(\sqrt{\frac{5d\log_e(2/\delta)}{i}} \vee \frac{5d\log_e(2/\delta)}{i} \right).$$

At time $t = \mathcal{T}_i + 1$, we use a change of measure to align our Meta-DP++ algorithm's prior $\mathcal{N}(\theta^{\mathrm{MDP}}_{i,\mathcal{T}_i+1}, \Sigma^{\mathrm{MDP}}_{i,\mathcal{T}_i+1})$ to $\mathcal{N}(\theta^{\mathrm{TS}}_{i,\mathcal{T}_i+1}, \Sigma^{\mathrm{MDP}}_{i,\mathcal{T}_i+1})$. Combining Lemma 3 and the fact that both policies offer the same prices in the random exploration periods, we know that $\Sigma^{\mathrm{TS}}_{i,\mathcal{T}_i+1}$ and $\Sigma^{\mathrm{MDP}}_{i,\mathcal{T}_i+1}$ are close with high probability for later epochs. However, it remains to bound the regret difference between the meta oracle's policy, which employs the prior $\mathcal{N}(\theta^{\mathrm{TS}}_{i,\mathcal{T}_i+1}, \Sigma^{\mathrm{TS}}_{i,\mathcal{T}_i+1})$, and our Meta-DP++ algorithm, which employs the prior $\mathcal{N}(\theta^{\mathrm{TS}}_{i,\mathcal{T}_i+1}, \Sigma^{\mathrm{MDP}}_{i,\mathcal{T}_i+1})$. We leverage importance sampling arguments from off-policy evaluation (Precup et al. 2000, Murphy et al. 2001) to bound this remaining term. Prior widening is instrumental in this last step, ensuring that our importance weights do not diverge.

Remark 6. Although our Meta-DP algorithm does not require prior widening, we widen our prior for our Meta-DP++ algorithm as described. This allows us to shave off some extra factors of the dimension d in our analysis by ensuring that the importance weights are well behaved postalignment. This is consistent with recent work by Hamidi and Bayati (2020), who show that Thompson sampling can, in general, incur a worst-case regret that scales exponentially in d unless it uses a widened posterior variance at each step. Furthermore, we observe (often significantly) improved empirical performance on both synthetic and real data sets by employing our Meta-DP++ algorithm compared with its nonwidened analog (see Section 5).

4.4. Additional Remarks

4.4.1. Hierarchical Model. An alternative heuristic to leverage shared structure is to use hierarchical Thompson sampling, maintaining a posterior on the shared prior and updating it after each epoch. In Online Appendix G.1, we compare the Meta-DP algorithm to a hierarchical approach; although the hierarchical algorithm outperforms prior-independent Thompson sampling by leveraging shared structure, we find that it still significantly underperforms compared with the Meta-DP algorithm for moderate-to-large values of N because of excessive exploration.

4.4.2. Knowledge of *N*, *T*. Our formulation assumes knowledge of N and T. However, this assumption can easily be removed using the well-known "doubling trick." In particular, we can initially fix any values N_0 and T_0 and iteratively double the length of the respective horizons; we refer the interested reader to Cesa-Bianchi and Lugosi (2006) for details. For the Meta-DP algorithm, we would simply continue to update the estimated prior mean; for the Meta-DP++ algorithm, we would need to also follow the prior widening schedule. It is easy to see that our regret bounds are preserved up to logarithmic terms under such an approach.

4.4.3. Overlapping Epochs. We model epochs as fully sequential for simplicity; if epochs overlap, we would need to additionally model a customer arrival process for each epoch. Our algorithms straightforwardly generalize to a setting in which arrivals are randomly distributed across overlapping epochs. In particular, both the Meta-DP algorithm and the Meta-DP++ algorithm can be modified to only use samples from the *initialization period* $t \in [\mathcal{T}_i]$ in each epoch for estimating the prior mean (note that our estimation of the prior covariance already only uses samples from initialization periods) without affecting the meta regret bounds and

analysis. Therefore, when epochs overlap, we update our estimate of the prior as soon as we see $\tilde{O}(1)$ customer responses for any product.

5. Numerical Experiments

We now validate our theoretical results by empirically comparing the performance of our proposed algorithms against prior-independent Thompson sampling (Agrawal and Goyal 2013). As discussed earlier, this approach ignores learning shared structure (the prior) across products and achieves $\tilde{O}(d^2 N\sqrt{T})$ meta regret (see Lemma 11 in Online Appendix C). When the prior covariance is unknown, we illustrate the benefits of prior widening by additionally comparing against a version of the Meta-DP++ algorithm that greedily uses the estimated covariance matrix (i.e., $\Sigma_i = \hat{\Sigma}_i$).

In addition to meta regret, we present results on Bayes regret (relative to the classical oracle) to illustrate that our transfer learning approach significantly increases performance under the standard metric. We perform numerical experiments on both synthetic data as well as a real data set on auto loans provided by the Columbia University Center for Pricing and Revenue Management.

A number of additional numerical results are presented in Online Appendix G, including comparison with a hierarchical Thompson sampling heuristic (G.1), examining the estimation error of the prior as a function of N (G.2), and results under a revenue metric (G.3).

5.1. Synthetic Data

We begin with the case in which the prior covariance Σ_* is known.

5.1.1. Parameters. We consider $n = 700$ products, each with a selling horizon of $T = 300$ periods. We set the feature dimension $d = 5$, the prior mean $\theta_* = [1.2 \times \mathbf{1}_d; -0.3 \times \mathbf{1}_d]^\top$, and the prior covariance $\Sigma_* = 0.2 \times I_{2d}$. In each epoch $i \in [N]$ and each round $t \in [T]$, each entry of the observed feature vector $x_{i,t}$ is drawn i.i.d. from the uniform distribution over $[0, 1/\sqrt{d}]^d$; note that this ensures the ℓ_2 norm of each feature vector is upper bounded by one. For each product $i \in [N]$, we randomly draw a demand parameter θ_i i.i.d. from the true prior $\mathcal{N}(\theta_*, \Sigma_*)$. The allowable prices lie in $(0, 5]$. Finally, the noise distribution is the standard normal distribution, that is, $\sigma = 1$.

5.1.2. Results. We plot the cumulative meta regret and Bayes regret of each algorithm, averaged over 20 random trials as a function of the number of epochs N (recall that each epoch lasts for T periods). The results are shown in Figure 1. Both algorithms are identical during the initial exploration epochs.

As expected, the prior-independent approach achieves meta regret that scales linearly in N because each epoch is treated independently. In contrast, the left panel of Figure 1 shows that Meta-DP achieves nearly zero meta regret after the exploration epochs as it has learned the prior.

The right panel of Figure 1 examines Bayes regret; note that even the meta-oracle achieves $O(N)$ Bayes regret (Theorem 1). However, the *slope* of Meta-DP closely matches that of the meta oracle after the initial exploration epochs; that is, we do not accrue additional regret (relative to the meta oracle) as N grows large. In contrast, the slope of prior-independent Thompson sampling is significantly larger, resulting in additional regret continually accruing as N grows large. In particular, when $N = 700$, the Bayes regret of prior independent Thompson Sampling is 39% larger than that of Meta-DP and 48% larger than that of the meta oracle. Thus, our approach of learning shared structure is particularly valuable in experiment-rich environments.

Figure 1. (Color online) Cumulative Meta Regret and Bayes Regret for Meta-DP and Prior-Independent Thompson Sampling

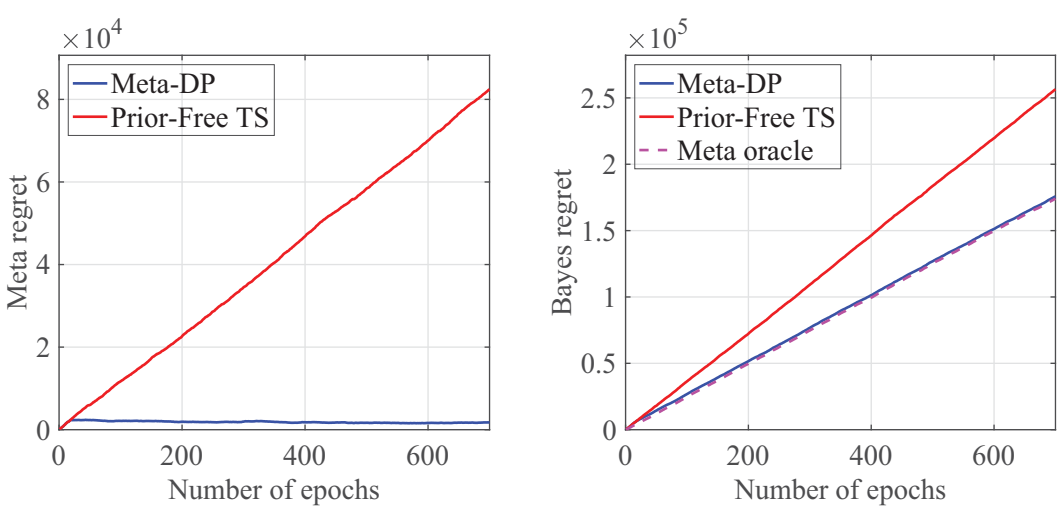
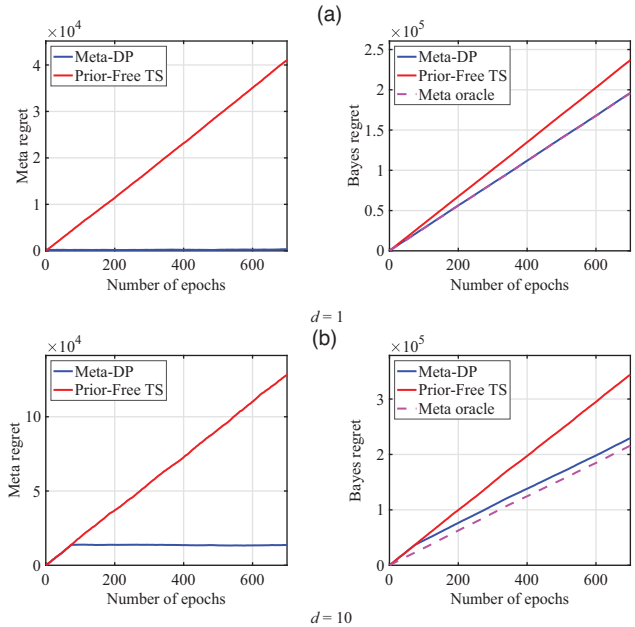


Figure 2. (Color online) Cumulative Meta Regret and Bayes Regret for Meta-DP and Prior-Independent Thompson Sampling for Different Values of the Feature Dimension d



Notes. (a) $d = 1$. (b) $d = 10$.

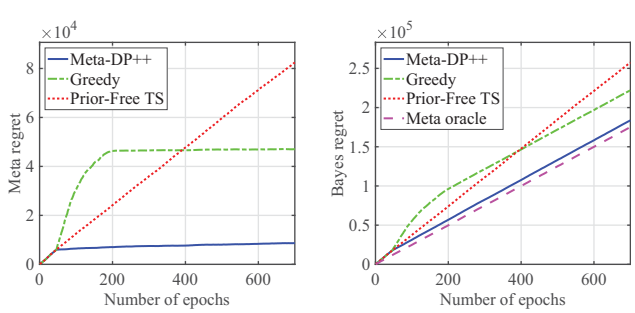
5.1.3. Varying the Feature Dimension *d*. We now explore how our results vary as we change the dimension of the observed features. Our previous results considered $d = 5$. We now additionally consider

1. No features, $d = 1$: We set $x_{i,t} = 1$ for all $i \in [N]$ and $t \in [T]$.

2. Many features, $d = 10$: Each entry of the observed feature vector $x_{i,t}$ is again drawn i.i.d. from the uniform distribution over $[0,1/\sqrt{d}]^d$ for all $i \in [N]$ and $t \in [T]$.

The results for both cases, averaged over 20 random trials, are shown in Figure 2, (a) and (b). Again, we see that Meta-DP substantially outperforms the prior-independent Thompson sampling algorithm in both meta and Bayes regret regardless of the choice of feature dimension d. Note that we require more exploration epochs when d is larger (because N_0 scales as d).

Figure 3. (Color online) Cumulative Meta Regret and Bayes Regret for Meta-DP++ and Benchmark Algorithms



5.1.4. Unknown Prior Covariance Σ_*. We now shift our attention to the Meta-DP++ algorithm and follow the same setup described earlier. To quantify the benefit of prior widening, we additionally consider a version of the Meta-DP++ algorithm that *greedily* uses the estimated covariance matrix, that is, $\Sigma_i = \hat{\Sigma}_i$. The results, averaged over 20 random trials, are shown in Figure 3. We see that the Meta-DP++ algorithm significantly outperforms both the prior-independent Thompson sampling algorithm as well as the nonwidened greedy benchmark in meta (left panel) and Bayes regret (right panel). Interestingly, the greedy approach performs significantly worse in earlier epochs after the initial exploration epochs (when it relies on a prior that is likely to be significantly misspecified); in later epochs, the greedy approach's slope begins to match that of Meta-DP++ as it starts learning the true prior. Thus, prior widening appears critical to ensure good performance on each pricing problem, particularly earlier ones, in which we should be careful not to over-rely on a prior that is likely to be significantly misspecified. The overall success of Meta-DP++ suggests that the price of not knowing the prior in advance is negligible in experiment-rich environments (large N).

5.2. Real Data on Online Auto Lending

We now turn to the online auto lending data set. This data set was first studied by Phillips et al. (2015) and subsequently used to evaluate dynamic pricing algorithms by Ban and Keskin (2021). We will follow a similar set of modeling assumptions.

The data set records all auto loan applications received by a major online lender in the United States from July 2002 through November 2004. It contains 208,085 loan applications. For each application, we observe some loan-specific features (e.g., date of application, the term and amount of loan requested, and the borrower's personal information), the lender's pricing decision (i.e., the monthly payment required of the borrower), and the resulting demand (i.e., whether this offer was accepted by the borrower). We refer the interested reader to Columbia University Center for Pricing and Revenue Management (Columbia 2015) for a detailed description of the data set.

5.2.1. Algorithms. We consider the setting in which both the prior mean and prior covariance are unknown. Thus, we compare the performance of our Meta-DP++ algorithm against that of prior-independent Thompson sampling, the ILSX algorithm proposed in Ban and Keskin (2021),[6] and the greedy version of Meta-DP++ that does not employ prior widening.

5.2.2. Products. We first define a set of related products. We segment loans by the borrower's state (there are 50 states), the term class of the loan (0–36, 37–48,

49–60, or more than 60 months), and the car type (new, used, or refinanced). The expected demand and loan decisions offered for each type of loan is likely different based on these attributes. We consider loans that share all three attributes as a single "product" offered by the online lender. We, thus, obtain a total of $n = 589$ unique products. The number of applicants in the data for each loan type determines T for each product; importantly, note that T is not identical across products.

5.2.3. Features. We use the feature selection results from Ban and Keskin (2021), which yields the following features: FICO score, the loan amount approved, prime rate, and the competitor's rate.

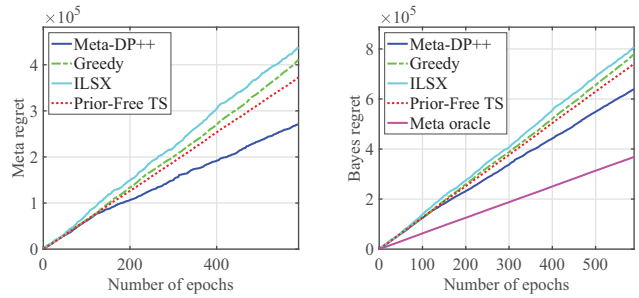
Remark 7. We use three categorical features (state, term of loan, and car type) to define $n = 589$ products, and we additionally have four numerical features per product. In contrast, the ILSX algorithm (Ban and Keskin 2021) sets $n = 1$ and encodes this information as product features; this results in a feature vector of dimension $4 + 50 + 4 + 3 = \Theta(d + N)$ because each possible value of the categorical feature is represented as a one-hot encoding. The resulting meta regret of ILSX, therefore, still grows superlinearly in N (unlike our proposed algorithms). Moreover, their demand model is less expressive compared with ours because it does not allow for different price elasticities by state/term/car type (see our earlier Remark 2 for discussion).

Remark 8. Following our model, we simulate each epoch sequentially. In reality, customers likely arrive randomly for each loan type at different points of time. We note that the Meta-DP algorithm only uses the initial sample from each epoch for estimating the prior mean, and thus, in principle, it can be adapted to a setting in which arrivals are randomly distributed across overlapping epochs as well (see discussion in Section 4.4).

5.2.4. Setup. Following the approach of Phillips et al. (2015) and Ban and Keskin (2021), we impute the price of a loan as the net present value of future payments (a function of the monthly payment, customer rate, and term approved; we refer the reader to the cited references for details). The allowable price range in our experiment is $[0, 30]$.

We note that, although we use a linear demand model, our responses are binary (i.e., whether a customer accepts the loan). This approach is common in the literature (see, e.g., Li et al. 2010). Besbes and Zeevi (2015) provide theoretical justification for this approach by showing that we may still converge to the optimal price despite the demand model being misspecified.

Figure 4. (Color online) Computational Results on a Real Data Set on Online Auto Loans



Note. Cumulative meta regret and Bayes regret for Meta-DP++ and benchmark algorithms.

Finally, unlike our model and analysis, the true distribution over loan demand parameters across products may not be a multivariate Gaussian. We use the entire data set to estimate each product's demand parameter and then fit a multivariate Gaussian prior over the empirical distribution of product demand parameters; our meta oracle uses this prior. However, our regret is evaluated with respect to the true data (i.e., our meta oracle may perform poorly in Bayes regret if the prior is far from a multivariate Gaussian). Thus, this experiment can provide a check on whether our algorithms (which seek to mimic the meta oracle) are robust to model misspecification of the prior.

5.2.5. Results. We average our results over 100 random permutations of the data. The results are shown in Figure 4. We first note that, despite potential misspecification of the prior's model class, the meta oracle (prior-dependent Thompson sampling) achieves much better Bayes regret (right panel) than all algorithms. This implies that the (potentially misspecified) shared prior across products is informative, and thus, leveraging shared structure may be valuable. Then, by design, our Meta-DP++ algorithm learns this shared structure, incurring meta regret that grows sublinearly in N (left panel). Consistent with our results on synthetic data, we see that the Meta-DP++ algorithm significantly outperforms the benchmark algorithms; this is true even though the multivariate Gaussian prior that we estimate may not be the true prior. This result suggests that our proposed algorithms may be robust to model misspecification of the prior.

6. Discussion and Conclusions

Firms are increasingly performing experimentation. This provides an opportunity for decision makers to learn not just within experiments, but also across experiments. In this paper, we consider the multiproduct dynamic pricing setting in which a decision maker must learn a sequence of related unknown parameters

through experimentation; we capture the relationship across these unknown parameters by imposing that they arise from a shared distribution (the prior). We propose meta-learning policies that efficiently learn both the shared distribution across experiments and the individual unknown parameters within experiments.

Our meta-learning approach can easily be adapted beyond dynamic pricing applications to classical multi-armed and contextual bandit problems as well. For instance, consider clinical trials, which were the original motivation for bandit problems (Thompson 1933, Lai and Robbins 1985). Many have argued the benefits of Bayesian clinical trials, which allow for the use of historical information and for synthesizing results of past relevant trials; for example, past clinical trials on the same disease may indicate that patients with certain biomarkers or concomitant medications are less likely to benefit from standard therapy. Such information can be encoded in a Bayesian prior to potentially allow for more informative clinical trials and improved treatment allocations to patients within the trial (see, e.g., Berry 2006, Anderer et al. 2019). Our meta-learning approach can inform how such priors are constructed. Importantly, prior widening gracefully transitions from an uninformative to an informative prior as we accrue data from more related clinical trials.

Our prior widening technique is inspired by the emerging literature studying prior misspecification in Thompson sampling. In general, adopting a more conservative prior allows Thompson sampling to still achieve the optimal theoretical guarantee, and a less conservative prior may cause failure to converge (Honda and Takemura 2014, Liu and Li 2015). However, the use of a conservative prior often results in poor empirical performance and can erode the benefit of using Thompson sampling over UCB and other prior-free approaches (see, e.g., Russo and Van Roy 2014, Bastani et al. 2021). We take the view that a successful implementation of Thompson sampling *requires* learning an appropriate prior and propose meta-learning policies to achieve this goal across a sequence of learning problems.

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Endnotes

[1] We provide some theoretical support for this finding because we show that limited prior misspecification does not affect the rate of convergence (e.g., when the prior covariance is known but the mean is unknown).

[2] We model epochs as fully sequential for simplicity; if epochs overlap, we need to additionally model a customer arrival process for each epoch. Our algorithms straightforwardly generalize for overlapping epochs; see remark in Section 4.4.

[3] Following the literature on Thompson sampling, we consider a multivariate Gaussian distribution because the posterior has a simple closed form, thereby admitting a tractable theoretical analysis. When implementing such an algorithm in practice, more complex distributions can be considered (e.g., see discussion in Russo et al. 2018).

[4] We use the term "meta oracle" to distinguish from the oracle in the classical formulation.

[5] When estimating the prior covariance, we cannot use an estimator of θ_i that uses all T observations from epoch i (as we do when estimating the prior mean). This is because the use of the learned prior from past epochs renders observations from later epochs nonindependent. We avoid this issue by restricting our estimator of θ_i to observations from the initialization periods in each epoch, $t \in [\mathcal{T}_i]$.

[6] We do not use a ℓ_1 penalty in our implementation because the "true support" identified by Ban and Keskin (2021) with all observations is precisely the set of features we consider.

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